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Direct Least-Squares Fitting of Ellipsoid to 3D Oriented Points for Point Cloud Processing

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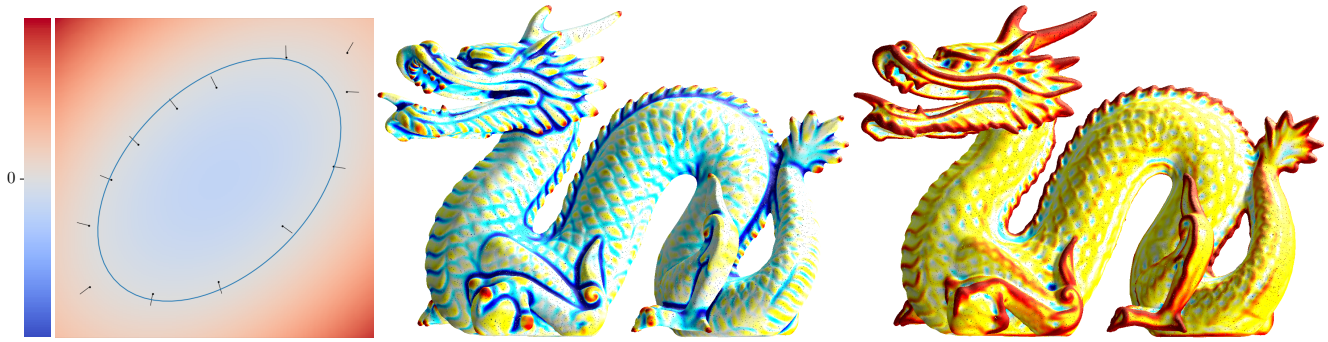

Figure 1: *Left: we propose a direct ellipsoid fitting algorithm that takes as input points with oriented normal vectors (in black) and computes the best fitting ellipsoid defined by the* 0*-isosurface (in blue) of a scalar field (color gradient). Middle and right: minimal and maximal principal curvature computed on a 3D point cloud using ellipsoid fitting.*

Abstract

We present a new algorithm for direct fitting of an ellipsoid to 3D points with oriented normal vectors. Locally fitting a canonical smooth surface to a set of scattered points is a fundamental operation in geometry processing. A fitted ellipsoid gives a direct access to second order differential properties that are important for analyzing points-sampled shapes. 3D acquisition techniques are now able to easily scan billions of points, which requires accurate and efficient methods to locally characterize point-sampled surfaces. The fitting method we present is efficient as it requires few space in memory and needs to traverse the neighborhood of a point only once to compute the ellipsoid.

1. Introduction

Point clouds are a widely used type of data that mainly arise from 3D acquisition using computer vision methods like Multi-View Stereo or depth sensors like Lidar. Their use is widespread across many domains such as robotics, retro-engineering and culture heritage to name a few. Unlike images, point clouds lack of regular structure, so standard signal processing tools cannot be directly used. Therefore, there is a need to develop new methods to process 3D point clouds for performing many different geometry processing tasks: registration, denoising, smoothing, resampling, meshing, etc.

In a typical processing pipeline, local surface approximation from scattered data points is a fundamental operation. Global registration [GMGP05] requires to compute local features to match two different point clouds sampling the same shape. Marching cubes [LC98] is a standard method to mesh a point cloud based on a local implicit approximation of the surface. Recent deep learning approaches [HLP*21, RRL23] still rely on hand-crafted features that usually include differential properties that are estimated locally.

In this work, we focus on the problem of fitting a canonical smooth surface on a small neighborhood around a point in an unstructured 3D point cloud. Unlike iterative methods like RANSAC [SWK07] and global methods like Poisson Surface Reconstruction [KBH06], we investigate *local* and *direct* surface approximation for efficiency purpose. In this way, direct fitting can be easily computed in parallel by GPU on large point clouds of over

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hundreds of millions of points, without the need to globally solve sparse but large systems.

Principal Component Analysis (PCA) is one of the most popular method that linearly approximate scattered points by fitting a plane [HDD*92]. Other algorithms exist to directly fit algebraic spheres to 3D points [Pra87, GG07, GGG08] with or without normal vectors. Fitting such isotropic shape gives access to the mean curvature of the sampled-surface [LCBM21] and has shown to be more robust to noise and more adapted to complex shapes in practice. However, sphere fitting cannot be used to directly estimate principal curvatures. To our knowledge, there is no algorithm to directly fit an ellipsoid to 3D points that come with oriented normal vectors, which is the contribution of this paper presented in Section 3. We also illustrate several applications of fitting ellipsoids to scanned 3D shapes in Section 4.

2. Related Work

We first give a state-of-the-art on the specific problem of direct ellipse and ellipsoid fitting in both 2D and 3D. We then review several methods that locally approximate a surface and that estimate differential properties in 3D point clouds. Global surface reconstruction methods [BTS*17] and other type of primitive fitting [KYZB19] are not discussed in this work.

Ellipse and ellipsoid fitting. Direct ellipse fitting in 2D usually amounts to minimize the sum of squared algebraic distances from the input points to a conic section [Zha97]. Existing methods mainly differ in the constraint applied on the parameters of the quadratic form to avoid the trivial null solution [Boo79, FPF99]. Extending these methods for 3D ellipsoid fitting is difficult because adding one dimension often leads to unconstrained least-squares problems. Nevertheless, it remains possible to fit specific ellipsoids, with a specific ratio of radii for instance [LG04].

Fitting 2D circles and 3D spheres is a simpler problem since there is less parameters in the model. The quadratic constraint that imposes the gradient to have a unit norm on the surface of the model leads to an accurate and efficient direct fitting algorithm [Pra87]. This algebraic sphere fitting is the basis of the Algebraic Point Set Surfaces (APPS) [GG07] used for 3D point cloud rendering. APSS also use another unconstrained direct sphere fitting algorithm taking into account globally oriented normal vectors [GGG08]. This latter approach has not been investigated for direct ellipsoid fitting to 3D oriented points.

Local surface approximation. Plane fitting using PCA is an efficient way to locally approximate a surface by a linear model. Many methods rely on this fundamental operation that can be seen as a tangent plane or normal vector estimation [HDD*92, PKG03]. Eigenvalues of the PCA are used for dimensionality selection [DMDV12] and classification [BL12].

Polynomial fitting of osculating jets over a PCA plane [CP05] estimate differential properties at any order at the cost of a large system to solve. More recently, the Wavejets [BDC18] fit a polynomial with a radial and angular decomposition.

Point Set Surfaces [ABCO*03] is another surface approximation framework based on the Moving Least-Squares [Lev98] that ensure that the surface is smooth everywhere. They have been originally developed using polynomial fitting over an estimated tangent plane, but many other variant exist [ÖGG09, CGBG13, RGRG15]. For instance, the APPS [GG07, GGG08] directly fit an algebraic sphere. The Algebraic Shape Operator [LCBM21] extend the algebraic sphere fitting with weighting kernel differentiation to estimate principal curvatures.

Integral invariant [PWHY09] link local integration of geometric quantities to differential invariants of the surface. Several curvature estimators have been analyzed from the integral invariant viewpoint [DM14, LCBM21].

Finally, many other curvature estimators exist that are based on various techniques: robust statistics [KSNS07], Voronoi diagram [MOG10], neural network [GKOM18], and corrected curvature measures computed on randomly generated triangles [LCL*23].

3. Method

Let $\mathcal{P} = \{\mathbf{p}_i, \mathbf{n}_i\}_{i=1\ldots n}$ be an input oriented point cloud with n 3D points $\mathbf{p}_i \in \mathbb{R}^3$ and their associated oriented normal vectors $\mathbf{n}_i \in \mathbb{R}^3, \|\mathbf{n}_i\| = 1$. We assume that $\mathcal{P}$ samples an unknown smooth regular 3D surface $\mathcal{S}$ that we want to estimate locally around each input point. Our objective is to fit an *algebraic ellipsoid* defined implicitly by a scalar field $f : \mathbb{R}^3 \to \mathbb{R}$,

$$f(\mathbf{x}) := u_c + \mathbf{u}_l^T \mathbf{x} + \mathbf{x}^T U_q \mathbf{x}. \tag{1}$$

The fitting is performed to 3D oriented points $\mathcal{N}_i \subset \mathcal{P}$ that corresponds to the local neighborhood of an input point $(\mathbf{p}_i, \mathbf{n}_i)$. In Equation 1, the scalar $u_c \in \mathbb{R}$, the vector $\mathbf{u}_l \in \mathbb{R}^3$, and the real square matrix $U_q \in \mathbb{R}^{3\times 3}$ are the unknown parameters of the ellipsoid. As illustrated in Figure 1, the 0-isosurface $\left\{\mathbf{x} \in \mathbb{R}^3, f(\mathbf{x}) = 0\right\}$ is the local estimate of the surface $\mathcal{S}$, and the subspace where f si negative (respectively positive) defines the interior (respectively the exterior) of the locally estimated surface.

3.1. Ellipsoid Fit

In order to fit u_c, $\mathbf{u}_l$ and U_q to the neighboring point set $\mathcal{N}_i$, we follow the APSS [GGG08] that fits an algebraic sphere to oriented points using a least-squares method. The fitting is performed in two steps that are detailed in the next sections.

3.1.1. Fitting the gradients to the normal vectors

We perform a weighted least-squares fit of $\mathbf{u}_l$ and U_q by minimizing the following sum of squared residuals between the gradient ∇f and the normal vectors

$$E_1(\mathbf{u}_l, U_q) := \frac{1}{2} \sum_{j \in \mathcal{N}_i} w_j \|\nabla f(\mathbf{p}_j) - \mathbf{n}_j\|^2$$

where w_j is a user-defined weight, and the gradient is defined from Equation 1 by

$$\nabla f(\mathbf{x}) := \mathbf{u}_l + \left(U_q + U_q^T\right)\mathbf{x}. \tag{2}$$

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Note that $j \in \mathcal{N}_i$ denotes now the indices of the neighbors of $\mathbf{p}_i$. We omit the detailed calculations and present the partial derivatives of E_1 as a first intermediate results of the least-squares fit

$$\frac{\partial E_1}{\partial U_q}(\mathbf{u}_l, U_q) = QA + AQ - C - C^T, \tag{3}$$

$$\frac{\partial E_1}{\partial \mathbf{u}_l}(\mathbf{u}_l, U_q) = \sum_j w_j\, \mathbf{u}_l + Q\sum_j w_j \mathbf{p}_j - \sum_j w_j \mathbf{n}_j, \tag{4}$$

where $Q = U_q + U_q^T$ and

$$A = \sum_j w_j \mathbf{p}_j \mathbf{p}_j^T - \frac{1}{\sum_j w_j}\sum_j w_j \mathbf{p}_j \sum_j w_j \mathbf{p}_j^T,$$

$$C = \sum_j w_j \mathbf{p}_j \mathbf{n}_j^T - \frac{1}{\sum_j w_j}\sum_j w_j \mathbf{p}_j \sum_j w_j \mathbf{n}_j^T.$$

Considering Equation 3 equal to zero leads to a special kind of Sylvester equation [BS72]

$$\boxed{QA + AQ = C + C^T}. \tag{5}$$

The corresponding small linear system is efficiently solved to obtain Q. Then, we can simply take $U_q = \frac{1}{2}Q$. Considering Equation 4 equal to zero directly leads to

$$\boxed{\mathbf{u}_l = \frac{1}{\sum_j w_j}\left(\sum_j w_j \mathbf{n}_j - 2U_q \sum_j w_j \mathbf{p}_j\right)}. \tag{6}$$

3.1.2. Fitting the scalar field to the points position

Knowing $\mathbf{u}_l$ and U_q from the previous step, we perform the least-squares fit of u_c by minimizing

$$E_0(u_c) := \frac{1}{2}\sum_{j \in \mathcal{N}_i} f(\mathbf{p}_i)^2.$$

We obtain

$$\frac{\partial E_0}{\partial u_c}(u_c) = u_c \sum_j w_j + \mathbf{u}_l^T \sum_j w_j \mathbf{p}_j + \sum_j w_j \mathbf{p}_j^T U_q \mathbf{p}_j$$

which, when equal to zero, gives

$$\boxed{u_c = -\frac{1}{\sum_j w_j}\left(\mathbf{u}_l^T \sum_j w_j \mathbf{p}_j + \sum_j w_j \mathbf{p}_j^T U_q \mathbf{p}_j\right)}. \tag{7}$$

3.2. Discussion

To understand Equation 5, we remark that A is the auto-covariance matrix of the points position (widely used to fit a plane to scattered data using PCA [HDD*92]), while C is the cross-covariance matrix between points position and normal vectors. Intuitively, Q represents a transformation matrix that matches points variance to normal vectors variance.

Our least-squares method to directly fit an algebraic ellipsoid to oriented points is similar to but more general than the APSS [GGG08] that fit an algebraic sphere. The main difference lies in the quadratic parameters of the scalar field. In Equation 1, U_q is a real matrix, while an algebraic sphere has a quadratic term of the form $u_q \mathbf{x}^T \mathbf{x}$, with $u_q \in \mathbb{R}$ a scalar. Equivalently, the algebraic sphere is an ellipsoid restricted to be isotropic, for which $U_q = u_q I$ with I the identity matrix.

From another point of view, we may see the algebraic ellipsoid of Equation 1 as a Taylor approximation of an implicit surface that is truncated at the second order. Similarly, the polynomial fitting of osculating jets [CP05] estimate a parametric surface as a truncated Taylor approximation over a tangent plane. Therefore, our method amounts to a sort of implicit 2-jet estimation.

Note that although we presented the fitting solutions as two steps 3.1.1 and 3.1.2 to find successively U_q, $\mathbf{u}_l$, and then u_c, the ellipsoid fit is performed by looking at the neighborhood $\mathcal{N}_i$ only once. To achieve this, we rewrite Equation 7 as

$$u_c = -\frac{1}{\sum_j w_j}\left(\mathbf{u}_l^T \sum_j w_j \mathbf{p}_j + \operatorname{trace}\left(\left(\sum_j w_j \mathbf{p}_j \mathbf{p}_j^T\right) U_q\right)\right).$$

With this small change, we avoid an additional neighbor query to compute the sum $\sum_j w_j \mathbf{p}_j^T U_q \mathbf{p}_j$ after having computed U_q, which may be costly in practice (even when using kd-trees or other accelerating data structures). To summarize, with one neighbor query, we compute the following sums: the scalar $\sum_j w_j$, the vectors $\sum_j w_j \mathbf{p}_j$ and $\sum_j w_j \mathbf{n}_j$, and the symmetric matrices $\sum_j w_j \mathbf{p}_j \mathbf{p}_j^T$ and $\sum_j w_j \mathbf{p}_j \mathbf{n}_j^T$, which takes a total of 19 scalar values in memory in 3D.

Finally, we emphasis that although U_q is not supposed to be symmetric by definition, the Equation 5 implies $Q = Q^T$ and thus $U_q = U_q^T$. It avoids the symmetrization step of the matrix of the quadratic form: $\mathbf{x}^T U_q \mathbf{x} = \mathbf{x}^T \frac{1}{2}(U_q + U_q^T)\mathbf{x}$.

4. Applications

To illustrate our ellipsoid fitting, this section explains two applications: curvature estimation and point cloud smoothing.

4.1. Curvature Estimation

In order to estimate the principal curvatures $\kappa_{\min}$ and $\kappa_{\max}$ of the smooth surface $\mathcal{S}$ sampled by $\mathcal{P}$ at a point $(\mathbf{p}_i, \mathbf{n}_i) \in \mathcal{P}$, we simply compute the curvatures of the locally fitted ellipsoid represented by the scalar field f of Equation 1. The curvatures of such implicit surface is obtained by performing the eigendecomposition of the shape operator

$$W := P^T \frac{\nabla^2 f}{\|\nabla f\|} P$$

where $P \in \mathbb{R}^{3\times 2}$ contains two orthonormal tangent vectors, ∇f is the gradient of Equation 2, and $\nabla^2 f$ is the Hessian matrix of f that is equal to

$$\nabla^2 f(\mathbf{x}) := U_q + U_q^T.$$

Contrary to many state-of-the-art methods for curvature estimation, our ellipsoid fitting method gives a direct access to second order differential properties. For instance, we avoid the use of weight kernel differentiation [LCBM21], larger system resolution [CP05, BDC18], and triangle generation [LCL*23]. Note that since U_q si symmetric, $\nabla^2 f = 2U_q$, and P is obtained using a Gram-Schmidt process applied on ∇f.

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Figure 1 (middle and right) shows the results of our principal curvature estimator using ellipsoid fitting on an acquired 3D point cloud.

4.2. Smoothing

To smooth a point cloud, we project each input points $\mathbf{p}_i$ onto their locally fitted ellipsoid, following the ideas of Point Set Surfaces [ABCO*03]. Given a point $\mathbf{x}$ and a fitted ellipsoid f (Equation 1), the projection $\bar{\mathbf{x}}$ of $\mathbf{x}$ onto the ellipsoid is necessarily along the gradient $\nabla f(\mathbf{x})$, which means there exists $t \in \mathbb{R}$ such that $\bar{\mathbf{x}} = \mathbf{x} + t\nabla f(\mathbf{x})$ and $f(\bar{\mathbf{x}}) = 0$. After injecting the first equation into the second one, we obtain the projection $\bar{\mathbf{x}}$ by solving for t the second-order polynomial

$$f(\mathbf{x}) + \left(\mathbf{u}_l^T \nabla f(\mathbf{x})\right) t + \left(\nabla f(\mathbf{x})^T U_q \nabla f(\mathbf{x})\right) t^2 = 0.$$

There is no solution if the ellipsoid is imaginary (if the fitting failed), or in very special cases (if $\mathbf{x}$ is the center of the ellipsoid for instance). In practice, theses cases do not happen because input points are close to their surface and there is enough points in the neighborhood so that the fitting succeeds. In Figure 2, a noisy point cloud is smoothed by projecting once each points onto ellipsoids fitted to their neighbors.

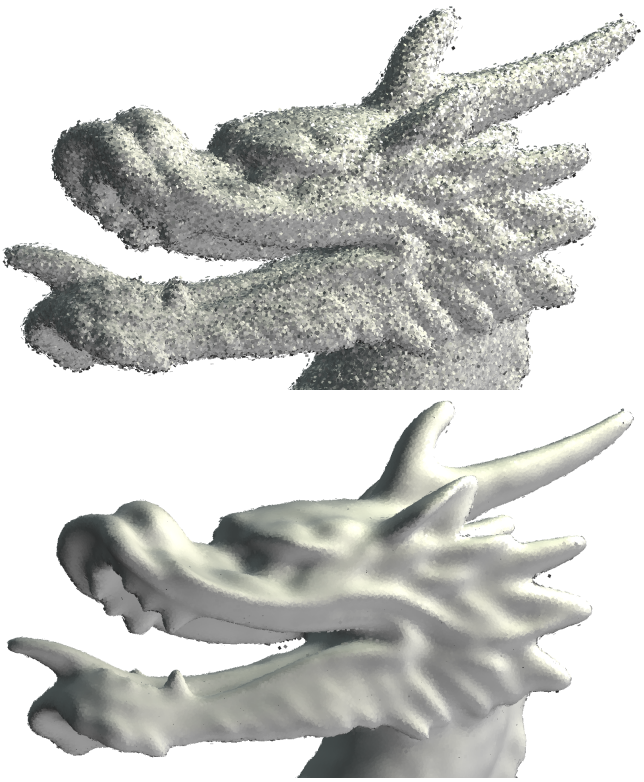

Figure 2: *Top: input point cloud with noisy points position and normal vectors. Bottom: smoothed point cloud where each point is projected onto their locally fitted ellipsoid.*

5. Conclusion and Future Work

We have presented a new algorithm to directly fit an ellipsoid to 3D points with normal vectors. Experiments have shown promising results for curvature estimation and more generally for geometry processing on unstructured point clouds. Several future work are considered apart from a quantitative evaluation and a comparison with state-of-the-art methods.

Requiring normal vectors with globally consistent orientation is an important limitation that could be addressed by developing an ellipsoid fitting algorithm for points with non-oriented normal vectors, as done for sphere fitting [CGBG13]. Analytical and experimental analysis of the impact of the prior normal estimation is also considered.

Moreover, we whish to perform a theoretical analysis of our method from the integral invariant view-point [PWHY09, DM14, LCBM21] to give convergence guarantees and to analyze the impact of noise.

Higher order differential estimation could be investigated using analytical or automatic differentiation of weighting kernel. Robustness to sharp features [ÖGG09] could also be an interesting improvement for robust geometry processing of scanned shapes.

Finally, we must assess the benefits of ellipsoid fitting to geometry processing by evaluating what it brings to several applications such as curvature estimation, meshing, and classification of unstructured point clouds.

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